

SYSC 5108 Exam Study

Assignment 4, Question 5

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Question Summary

We have a two-state, two-action reinforcement-learning problem:

State	Action	Next state	Probability	Reward
0	0	0	1	-1
0	1	1	1	1
1	0	1	1	1
1	1	0	1	-1

Assume:

$$\gamma = 0.1, \quad \pi(1 \mid 0) = \pi(0 \mid 1) = 0.6.$$

The question asks for:

- (a) ε for the ε -greedy policy,
- (b) the state-value and action-value functions under the given policy,
- (c) the optimal policy and optimal values,
- (d) Monte Carlo estimates from two generated trajectories,
- (e) off-policy TD control (Q-learning),
- (f) on-policy TD control (SARSA).

Course and Textbook Cross-References

- **Chapter 6 slides, Value Function and Solving Bellman Equation:** use Bellman equations for $v_\pi(s)$ and $q_\pi(s, a)$.
- **Chapter 6 slides, ε -greedy policy:** the greedy action gets probability $1 - \varepsilon + \varepsilon/|\mathcal{A}|$.
- **Chapter 6 slides, Monte Carlo:** update values from returns G_t .
- **Chapter 6 slides, Q-learning:**

$$q(s, a) \leftarrow q(s, a) + \alpha \left[r + \gamma \max_{a'} q(s', a') - q(s, a) \right].$$

- **Chapter 6 slides, SARSA:**

$$q(s, a) \leftarrow q(s, a) + \alpha \left[r + \gamma q(s', a') - q(s, a) \right].$$

Part (a): Find ε

There are two actions, so an ε -greedy policy gives the greedy action probability

$$1 - \varepsilon + \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2}.$$

The problem states

$$\pi(1 \mid 0) = \pi(0 \mid 1) = 0.6,$$

and these are the greedy-action probabilities in states 0 and 1, respectively. Therefore

$$1 - \frac{\varepsilon}{2} = 0.6 \quad \Rightarrow \quad \frac{\varepsilon}{2} = 0.4 \quad \Rightarrow \quad \varepsilon = 0.8.$$

$\varepsilon = 0.8$

Part (b): Find V_π and Q_π

Under the given policy:

$$\begin{aligned}\pi(0 \mid 0) &= 0.4, & \pi(1 \mid 0) &= 0.6, \\ \pi(0 \mid 1) &= 0.6, & \pi(1 \mid 1) &= 0.4.\end{aligned}$$

The Bellman equations are

$$\begin{aligned}V(0) &= 0.4[-1 + 0.1V(0)] + 0.6[1 + 0.1V(1)], \\ V(1) &= 0.6[1 + 0.1V(1)] + 0.4[-1 + 0.1V(0)].\end{aligned}$$

By symmetry, let

$$V(0) = V(1) = v.$$

Then

$$\begin{aligned}v &= 0.4(-1 + 0.1v) + 0.6(1 + 0.1v) \\ &= -0.4 + 0.04v + 0.6 + 0.06v \\ &= 0.2 + 0.1v.\end{aligned}$$

So

$$0.9v = 0.2 \quad \Rightarrow \quad v = \frac{2}{9} \approx 0.2222.$$

Therefore

$V(0) = V(1) = \frac{2}{9} \approx 0.2222.$

Now compute the action values:

$$\begin{aligned}Q(0, 0) &= -1 + 0.1V(0) = -1 + \frac{1}{45} = -\frac{44}{45} \approx -0.9778, \\ Q(0, 1) &= 1 + 0.1V(1) = 1 + \frac{1}{45} = \frac{46}{45} \approx 1.0222, \\ Q(1, 0) &= 1 + 0.1V(1) = \frac{46}{45} \approx 1.0222, \\ Q(1, 1) &= -1 + 0.1V(0) = -\frac{44}{45} \approx -0.9778.\end{aligned}$$

$$\begin{aligned}Q(0, 0) &= -0.9778, & Q(0, 1) &= 1.0222, \\ Q(1, 0) &= 1.0222, & Q(1, 1) &= -0.9778.\end{aligned}$$

Part (c): Find the Optimal Policy

The optimal Bellman equations are

$$V^*(0) = \max\{-1 + 0.1V^*(0), 1 + 0.1V^*(1)\},$$

$$V^*(1) = \max\{1 + 0.1V^*(1), -1 + 0.1V^*(0)\}.$$

Clearly the better action in state 0 is action 1, and the better action in state 1 is action 0. Thus

$$\boxed{\pi^*(0) = 1, \quad \pi^*(1) = 0}.$$

Under this policy,

$$V^*(1) = 1 + 0.1V^*(1) \quad \Rightarrow \quad 0.9V^*(1) = 1 \quad \Rightarrow \quad V^*(1) = \frac{10}{9}.$$

Then

$$V^*(0) = 1 + 0.1V^*(1) = 1 + \frac{1}{9} = \frac{10}{9}.$$

$$\boxed{V^*(0) = V^*(1) = \frac{10}{9} \approx 1.1111}.$$

Part (d): Monte Carlo Estimates

Use the random numbers in order:

$$0.94, 0.27, 0.30, 0.18, 0.75, 0.35, \dots$$

Action-selection convention under the given ε -greedy policy:

- in state 0, choose action 1 if $r \leq 0.6$, else action 0,
- in state 1, choose action 0 if $r \leq 0.6$, else action 1.

Trajectory starting from $s_0 = 0$

$$0 \xrightarrow[a=0]{-1} 0 \xrightarrow[a=1]{1} 1 \xrightarrow[a=0]{1} 1.$$

This uses random numbers 0.94, 0.27, 0.30.

Trajectory starting from $s_0 = 1$

$$1 \xrightarrow[a=0]{1} 1 \xrightarrow[a=1]{-1} 0 \xrightarrow[a=1]{1} 1.$$

This uses random numbers 0.18, 0.75, 0.35.

Returns

For the first trajectory:

$$\begin{aligned}G_0^{(1)} &= -1 + 0.1(1) + 0.1^2(1) = -0.89, \\G_1^{(1)} &= 1 + 0.1(1) = 1.1, \\G_2^{(1)} &= 1.\end{aligned}$$

For the second trajectory:

$$\begin{aligned}G_0^{(2)} &= 1 + 0.1(-1) + 0.1^2(1) = 0.91, \\G_1^{(2)} &= -1 + 0.1(1) = -0.9, \\G_2^{(2)} &= 1.\end{aligned}$$

State-value estimates

Using every-visit incremental Monte Carlo with $\alpha = 0.6$ and initial values equal to 0,

$$\boxed{V(0) \approx 0.77856, \quad V(1) \approx -0.22560}.$$

Action-value estimates

Again using every-visit incremental Monte Carlo with $\alpha = 0.6$:

$$\boxed{Q(0,0) = -0.534, \quad Q(0,1) = 0.864, \quad Q(1,0) = 0.786, \quad Q(1,1) = -0.540}.$$

Part (e): Off-Policy TD Control (Q-learning)

Use

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right],$$

with $\alpha = 0.6$, starting from

$$Q(s, a) = 0 \quad \text{for all } (s, a).$$

For the final step of each trajectory, treat the endpoint at $t = 3$ as terminal, so the target is just the immediate reward.

After processing the two trajectories from part (d), the final Q-values are:

$$\boxed{Q(0,0) = -0.600, \quad Q(0,1) = 0.840, \quad Q(1,0) = 0.876, \quad Q(1,1) = -0.564}.$$

Part (f): On-Policy TD Control (SARSA)

Use

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma Q(s', a') - Q(s, a) \right].$$

Start with

$$Q(s, a) = 0 \quad \text{for all } (s, a),$$

and initial policy

$$\pi(1 \mid 0) = \pi(0 \mid 1) = 0.5.$$

Using the same random sequence from the beginning,

$$0.94, 0.27, 0.30, 0.18, 0.75, \dots$$

the trajectory is

$$0 \xrightarrow[a=0]{-1} 0 \xrightarrow[a=1]{1} 1 \xrightarrow[a=0]{1} 1 \xrightarrow[a=0]{1} 1 \xrightarrow[a=1]{-1} 0.$$

Applying SARSA step by step gives:

$$Q(0,0) = -0.600, \quad Q(0,1) = 0.600, \quad Q(1,0) = 0.840, \quad Q(1,1) = -0.600.$$

Final Answer

$$\varepsilon = 0.8$$

$$V(0) = V(1) = \frac{2}{9} \approx 0.2222$$

$$\begin{aligned} Q(0,0) &= -0.9778, & Q(0,1) &= 1.0222, \\ Q(1,0) &= 1.0222, & Q(1,1) &= -0.9778 \end{aligned}$$

$$\pi^*(0) = 1, \quad \pi^*(1) = 0$$

$$V^*(0) = V^*(1) = \frac{10}{9} \approx 1.1111$$

$$\text{MC: } V(0) \approx 0.77856, \quad V(1) \approx -0.22560$$

$$\text{MC } Q : Q(0,0) = -0.534, \quad Q(0,1) = 0.864, \quad Q(1,0) = 0.786, \quad Q(1,1) = -0.540$$

$$\text{Q-learning: } Q(0,0) = -0.600, \quad Q(0,1) = 0.840, \quad Q(1,0) = 0.876, \quad Q(1,1) = -0.564$$

$$\text{SARSA: } Q(0,0) = -0.600, \quad Q(0,1) = 0.600, \quad Q(1,0) = 0.840, \quad Q(1,1) = -0.600$$

Exam Notes

- In this problem, the transitions are deterministic. The only randomness is from the behavior policy.
- For ε -greedy with two actions, the greedy action gets probability $1 - \varepsilon/2$, not $1 - \varepsilon$.
- Q-learning uses the greedy target $\max_{a'} Q(s', a')$, while SARSA uses the actually selected next action a' .
- Be explicit about how the random numbers are mapped to actions. That is where most bookkeeping mistakes happen.